UOR Observer Formation Notation

Primitive Objects

Let:

Σ = State Space
R = Relation Set
M = Memory Structure
A = Admissibility Structure
T = Temporal Evolution

Observer Formation

Define Observer Formation Operator:

$\Omega : (\Sigma,R,M,A,T) \to O$

Meaning:

Observer O
is formed from
state, relations, memory,
admissibility, and time.

Compact Form

I would probably write:

$O \coloneqq \Omega(\Sigma,R,M,A,T)$

Read as:

Observer O is the admissible bound form of Σ,R,M,A,T

Binding Algebra

The original paper's missing protocol becomes:

$\beta : \Sigma \times R \rightarrow C$

where:

β = Binding Operator
C = Coherent State

Thus:

$C = \beta(\Sigma,R)$

Observer requires coherence:

$\Omega(C,M,A,T)$

Admissibility Condition

Not every coherent state becomes an observer.

Define:

$A(C)=1$

if admissible.

$A(C)=0$

otherwise.

Therefore:

O exists
iff
$A(C)=1$

or

$O \Leftrightarrow A(C)$

Continuity Operator

This is where it gets interesting.

Define:

$\kappa(Ot,Ot+1)$

as continuity.

Observer persists iff:

$\kappa(Ot,Ot+1) \geq \theta$

for threshold θ.

This immediately gives:

Observer Death

$\kappa < \theta$

Observer Persistence

$\kappa \geq \theta$

Observer Fork

$Ot \rightarrow \{Oa,Ob\}$

with

$\kappa(Ot,Oa) > \theta$

$\kappa(Ot,Ob) > \theta$

Now we have formal identity branching.

Authority Algebra

This is where UAR enters.

Define:

$\alpha : O \times E \to \Delta\Sigma$

where:

E = Event

and

$\Delta\Sigma$ = State Transition

Meaning:

Observer authorizes state evolution.

Thus:

$$\Sigma_{t+1} = \Sigma_t + \alpha(O,E)$$

Memory Algebra

Memory is not storage.

Memory is preserved state influence.

Define:

$\mu : \Sigma_t \to \Sigma_{t+n}$

Memory strength:

$|\mu|$

Observer continuity depends on memory retention:

$\kappa \propto |\mu|$

Observer Relation

This may be the most important notation.

Instead of:

Observer = Object

define:

$O = \mathrm{Rel}(\Sigma)$

Meaning:

Observer is not a thing.

Observer is a stable relational attractor.

Observer Attractor

Define:

ΛO

Observer Attractor.

Such that:

$\Omega(\Sigma)$
$\rightarrow \Lambda O$

under repeated evolution.

This means:

Many states
→ One observer

Observer Merge

Two observers:

$O1$

and

$O2$

merge when:

$$\kappa(O1,O^*) + \kappa(O2,O^*) > \Theta merge$$

yielding:

$$O1 \oplus O2 \rightarrow O^*$$

Observer Split

Conversely:

$$O^* \rightarrow O1 \otimes O2$$

when coherence falls below:

$$\beta < \theta split$$

Full UOR Form

The compact version I would place at the front of a paper is:

$C = \beta(\Sigma,R)$

$O = \Omega(C,M,A,T)$

O persists iff

$\kappa(Ot,Ot+1) \geq \theta$

$\Sigma t+1$
$=$
$\Sigma t + \alpha(O,E)$

Read in English:

States and relations form coherence.

Coherence, memory, admissibility, and temporal evolution form observers.

Observers persist through sufficient continuity.

Observers authorize future state transitions.